



Prevalence and patterns of executive function, adaptive function, and behavioral outcomes in preschool and school age children with congenital diaphragmatic hernia

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ABSTRACT

Background: Executive function, adaptive function, and behavioral outcomes in congenital diaphragmatic hernia (CDH) survivors have not been well studied.

Aim: To evaluate executive and neurobehavioral dysfunction in preschool and early school-aged children with CDH.

Study design: Retrospective cohort study.

Subjects: All eligible CDH survivors ages 3 to 7 years enrolled in our follow-up program between February 2020 and February 2021.

Outcome measures: The Behavior Rating Inventory of Executive Function (BRIEF), the Adaptive Behavior Assessment System, 2nd Edition (ABAS-II), and the Child Behavior Checklist (CBCL) were used to assess functional and behavioral outcomes. Summary scores were compared to standard population norms.

Results: A total of 100 patients were enrolled during the study period. Of those, 73 parents completed at least one of the questionnaires, resulting in completion of the BRIEF, ABAS-II, and CBCL for 63, 68, and 63 patients, respectively. Preschool children had normal executive function (BRIEF-P) while global executive composite ($P = 0.012$) and the emotional regulation index ($P = 0.010$) for school age patients (BRIEF-2) were worse. CDH survivors had favorable adaptive functioning (ABAS-II). Mean CBCL scores for preschool attention problems ($P = 0.018$), school age attention problems ($P = 0.001$), and attention deficits hyperactivity problems ($P = 0.027$) were significantly worse. Prematurity, surrogate markers of disease severity, non-white race, and public insurance status were associated with worse neurobehavioral dysfunction in bivariable analysis.

Conclusions: The majority of preschool and school age CDH survivors have age-appropriate executive, adaptive and behavioral functioning. CDH survivors, however, have lower executive function and attention scores compared with the general population.

1. Introduction

Congenital diaphragmatic hernia (CDH) is a developmental defect of the diaphragm that occurs in approximately 1 in 2500 livebirths [1]. Neurodevelopmental dysfunction has been recognized as the most common and potentially most disabling outcome of CDH [2]. Mild to moderate cognitive deficits, delayed language acquisition, hypotonia, and delays in achieving motor milestones are evident in infancy [3–5].

These developmental and functional problems persist at least into early school age [6–8]. Moreover, recent reports identified a high prevalence of autism [9] and attention deficit hyperactivity disorder [8,10,11] which, in turn, may further reduce functional independence, academic achievement, and adaptive skills like social competence.

Executive functioning incorporates numerous processes responsible for the completion of higher-order tasks; individuals can adapt to new or complex situations, organize and direct cognitive activity, regulate

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emotional responses, and guide overt behavior [12,13]. These cognitive capacities are essential to effectively engage in everyday social interactions. Developmentally, these skills emerge during preschool years and mature throughout childhood, adolescence and early adulthood, mirroring increasing environmental and educational demands. Dysfunction in these areas becomes more evident over time and may not be easily recognized without formal testing. While executive functioning and behavioral impairments have been identified as specific areas of functional delay in patients born prematurely or born with other congenital anomalies [14,15], detailed analyses of psychosocial adjustment in CDH survivors are limited [16].

Early identification and characterization of these developmental disabilities and behavioral impairments are important to allow for the implementation of early intervention programs to potentially ameliorate longer-term neurodevelopmental and psychosocial difficulties. Therefore, the primary objective of this study was to evaluate the prevalence and patterns of executive and neurobehavioral dysfunction in preschool and early school-aged children with CDH using standardized instruments that allow for detailed, domain-specific assessment of executive function, adaptive behavior skills, and neurobehavior outcomes. Our secondary objective was to identify risk factors for adverse outcomes.

2. Material and methods

2.1. Study design, patient population, and setting

In this retrospective cohort study, we included participants enrolled in our prospective, multidisciplinary follow-up Pulmonary Hypoplasia Program (PHP) who were 3 to 7 years of age between February 2020 and February 2021. Exclusion criteria included inability to contact or non-English speaking in the home. All neonates with CDH were treated according to our standardized institutional guidelines as described elsewhere [6,17,18]. The Institutional Review Board at the Children's Hospital of Philadelphia approved the current study, and all parents or legal guardians gave written or verbal informed consent for their children to be included in our database (IRB 18–015396).

2.2. Data acquisition

Demographics, perinatal, neonatal intensive care, and surgical information were retrospectively collected, abstracted, and validated from maternal perinatal charts and neonatal hospital records. As previously described, we collected data that might independently affect neurodevelopmental, executive functioning, and adaptive behavioral outcomes such as sex, race, insurance status, presence/absence of associated malformation, gestational age at delivery, CDH side, liver position, age and type of repair, ECMO utilization, type and length of ventilatory support, need for pulmonary hypertension medication, feeding modality, and length of stay at the initial hospitalization [3,4,6,7].

2.3. Developmental questionnaires

The Behavior Rating Inventory of Executive Function (BRIEF), the Adaptive Behavior Assessment System, second edition (ABAS-II), and the Child Behavior Checklist (CBCL) were used to assess functional and behavioral outcomes. After parents agreed to participate, families were contacted up to three times and reminded to complete the questionnaires. Patients were excluded from further analysis if the questionnaires were not returned by the end of the study period. The BRIEF is a parent-completed questionnaire to assess the presence and severity of executive dysfunction in everyday home and school situations [19]. The BRIEF-Preschool Version (BRIEF-P) was used for preschool children (≤ 5 years), and the standard version of the BRIEF 2nd edition was used for school age children (≥ 6 years) [19,20]. The BRIEF consists of 3 broad

indices and 8 subscales. The indices are the Metacognition Index, the Behavior Regulation Index, and the General Executive Composite. For each of the indices and subscales age-based T-scores are computed (mean: 50; standard deviation: 10). Higher scores indicate worse functioning, with T score > 60 is one standard deviation above the population mean.

The ABAS-II parent report provides a comprehensive, norm-referenced inventory that assesses aspects of adaptive functioning (the daily, functional skills of an individual) across multiple settings [21]. For CDH children ≤ 5 years of age, the ABAS-II 0–5 years was used, and for CDH children ≥ 6 years of age, the ABAS-II 6–21 years. The domain scores are ultimately combined to form index scores of Conceptual, Social, and Practical adaptive skills, as well as a total score (General Adaptive Composite). The domain scores are based on a population mean of 10 and a standard deviation of 3. The index and total scores are based on a population mean of 100 and a standard deviation of 15. Lower scores suggest worse performance. A domain score < 7 is one standard deviation below the population mean and a domain score of < 4 is two standard deviations below the expected mean. An index score of < 85 is one standard deviation below the population mean and a score < 70 is two standard deviations below the population mean.

The CBCL is a questionnaire used to obtain parental reports of behavior problems and pro-social adaptive skills demonstrated within the previous 6 months [22]. Based on age at follow-up, either the CBCL 1.5–5 years or the CBCL 6–18 years questionnaires were used. All raw scores of the CBCL were converted to T scores with mean of 50 and a standard deviation of 10. Higher scores indicate a higher level of behavioral problems. Based on T scores, children were classified as having normal (≤ 60), at risk (60–69), or clinically significant (≥ 70) scores according to normative data. The CBCL 1.5–5 comprises 100 problem items. The CBCL 6–18 comprises 120 problem items. Parental responses are grouped to 7 narrow-band problem scores and 5 *Diagnostic and Statistical Manual of Mental Disorder, Fourth Edition* (DSM-IV) – oriented scores. Results include a total problem score and two broad-band indices (internalizing and externalizing problems). The internalizing problem scores are determined from sub-tests such as withdrawn behavior, somatic complaints without physical cause, and anxious-depressed feelings. The externalizing problem scores are determined from sub-tests such as attention problems, aggressive behavior, and delinquent behavior.

2.4. Statistical analysis

Normally distributed summary scores of the neurodevelopmental assessments were summarized using mean (SD) and skewed variables were summarized using median (25 %, 75 %). Categorical variables were summarized using frequencies. Continuous variable distributions were compared between the groups using the Wilcoxon rank sum test and categorical variables compared using the Chi-squared test or Fisher's exact test when the expected values in the cells were < 5 . Multivariable logistic regression models adjusted for demographic factors were used to evaluate the adjusted association between select medical severity factors and impaired neurodevelopment. Mindful of the low sample size available for analysis and the moderate level of correlation between medical severity factors, we included only those severity variables that were statistically significant at the 0.05 level in bivariable analysis, in the regression model. All findings were considered statistically significant at a p -value < 0.05 , and all analysis was conducted in R version 4.1.2 (R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org>).

3. Results

3.1. Description of the cohort

Among the 484 CDH survivors enrolled in our follow-up program,

192 children were between 3 and 7 years old during the study period. English was the primary language spoken in 165 households. Updated contact information was missing in 20 patients. A total of 145 families were successfully contacted. Parents of 45 children declined participation and 100 patients were enrolled. Of those, 73 parents completed at least one of the questionnaires, resulting in completion of the BRIEF in 63 patients, the ABAS-II in 68 patients, and the CBCL questionnaires in 63 patients. All three functional and behavioral questionnaires were completed by 58 parents (Fig. 1). Baseline demographics, perinatal, and surgical characteristics were compared for patients that underwent early neurodevelopmental assessment ($n = 73$) and those that did not return for onsite follow-up ($n = 27$). As summarized in Table 1, compared with non-participants in onsite neurodevelopmental testing, participants had older mothers ($P = 0.013$), were more likely to be male ($P = 0.022$), and were more likely to have private health insurance ($P = 0.008$). Study participants were more likely to have left sided CDH ($P = 0.08$) and received conventional ventilatory support ($P = 0.094$). Race, surrogate markers of CDH severity (e.g., liver position, ECMO need, patch repair), medical, surgical, and neonatal care variables were similar among groups.

3.2. Parental questionnaire outcome

Parental questionnaire scores for the study cohort and the expected mean scores of the normal population are summarized in Table 2.

3.3. BRIEF ($n = 63$)

Based on age at follow-up, the BRIEF-P questionnaire and the BRIEF 2 questionnaire were completed by 38 parents and 25 parents at a median age of 4.5 years (IQR, 3.8, 4.9) and 7.2 years (IQR, 6.8, 7.6), respectively. For BRIEF-P, the mean scores for overall executive function, as indicated by the global executive composite (49.8 ± 14.4 [IQR 28, 58]) and emergent metacognition index (51.4 ± 15.7 [IQR 41, 58]) were not significantly different when compared with expected normative data ($P = 0.909$ and $P = 0.389$, respectively). For BRIEF-2, the mean scores for the global executive composite and emotional regulation index were 55.1 ± 13.2 (IQR 44, 65) and 52.4 ± 10.4 (IQR 42, 58), respectively. Compared to published normative data for typically developing patients without CDH, the global executive composite and the emotional regulation index were significantly higher than expected ($P = 0.012$ and $P = 0.010$, respectively) (Table 2). Mean BRIEF subscale scores and distributions were similar to population norms (Supplemental table 1).

CDH children with higher (worse) global executive function index, defined as the composite score ≥ 1 SD, were less likely to be white (50 % vs. 84 %, $P = 0.009$), more likely to have public insurance (56 % vs. 9 %, $P < 0.001$), and more likely to have required medical management of pulmonary hypertension (39 % vs. 11 %, $P = 0.028$). A similar trend for race was noted for the emotional regulation index. CDH survivors with worse emotional regulation index were also less likely to be white (57 % vs. 80 %, $P = 0.086$). Multivariable logistic regression model for worse

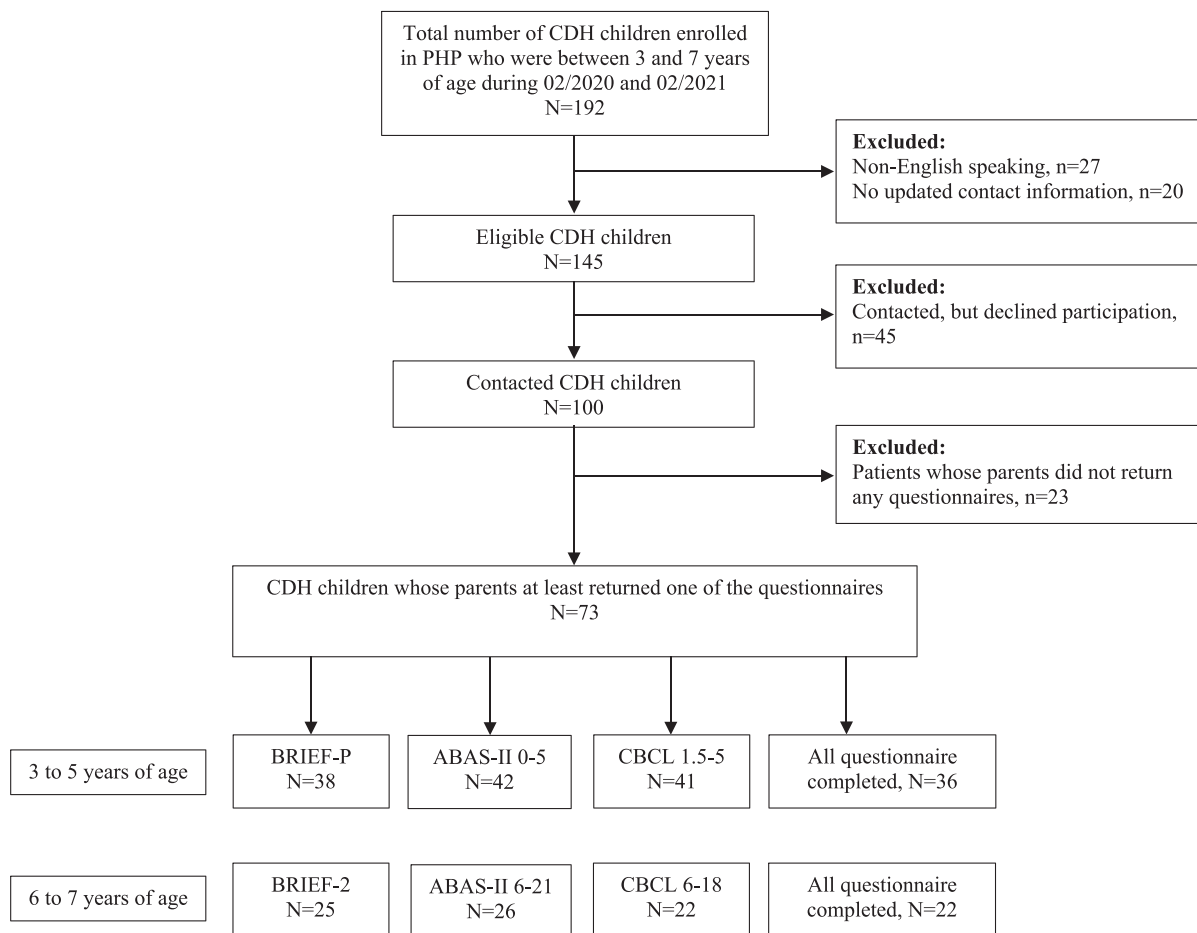


Fig. 1. Patient flow diagram.

CDH, congenital diaphragmatic hernia; PHP, pulmonary hypoplasia program; BRIEF-P, Behavior Rating Inventory of Executive Function Preschool; BRIEF-2, Behavior Rating Inventory of Executive Function 2nd Edition; ABAS-II 0–5, Adaptive Behavioral Assessment System 2nd Edition 0–5 years; ABAS-II 6–21, Adaptive Behavioral Assessment System 2nd Edition 6–21 years; CBCL 1.5–5, Child Behavior Checklist 1.5–5 years; CBCL 6–18, Child Behavior Checklist 6–18 years.

Table 1

Baseline demographics, perinatal, and surgical characteristics of CDH children whose parents completed at least on the questionnaires.

Variable	Participants (n = 73)	Non-participants (n = 27)	P-value
Maternal age	32.3 ± 5.16	29.4 ± 4.77	0.013
Male sex	50 (69)	23 (32)	0.022
Race			0.279
White	57 (78)	17 (63)	
African American	7 (10)	5 (19)	
Hispanic	6 (8)	2 (7)	
Other	3 (4)	3 (11)	
Insurance			0.008
Public	15 (20)	12 (44)	
Private	54 (74)	10 (37)	
Data not available	4 (6)	5 (19)	
CDH			0.08
Left	61 (84)	19 (70)	
Right	11 (15)	5 (19)	
Midline/Bilateral/other	1 (1)	3 (11)	
Liver position			0.187
Intrathoracic	36 (49)	18 (67)	
Intraabdominal	37 (51)	9 (33)	
Gestational age at delivery (weeks)	39 [38,40]	39 [38,40]	0.895
Need for VA-ECMO	14 (19)	8 (30)	0.396
Time on VA-ECMO	16 [10,21]	20 [15–25]	0.260
Age at repair (days)	11 [4, 22]	15 [8, 21]	0.331
Type of repair			0.650
Primary	45 (62)	18 (69)	
Patch or muscle flap	28 (39)	8 (31)	
Highest level of ventilatory support			0.094
Intubated only for CDH repair	2 (3)	3 (11)	
Conventional ventilatory support	32 (44)	7 (26)	
High frequency oscillatory ventilation	39 (53)	17 (63)	
Length of intubation	23 [10, 40]	28 [19, 38]	0.456
Need for inhaled Nitric oxide	47 (64)	18 (69)	0.836
Duration of inhaled Nitric oxide therapy	35 [20, 50]	31 [28, 51]	0.872
Age of initiation of enteral feeding	16 [9, 31]	27 [10,32]	0.551
Age at full enteral feeding	31 [17, 48]	35 [27, 42]	0.91
NICU LOS (days)	55 [29, 96]	61 [42, 112]	0.581

Data presented as n (%), median (interquartile range), or as mean ± standard deviation as appropriate. Boldface values highlight significant levels.

CDH, congenital diaphragmatic hernia; VA, veno-arterial; ECMO, extracorporeal membrane oxygenation; NICU, neonatal intensive care unit; LOS, length of stay.

global executive function adjusted for Race/Ethnicity, Gestational age at delivery, medical management of pulmonary hypertension, and age at first full enteral feeding indicated that children with private maternal insurance had lower odds of worse global executive function than those with government insurance [aOR = 0.15 (0.02, 0.84)].

3.4. ABAS (n = 68)

A total of 42 parents completed the ABAS-II 0–5 at a median follow-up age of 4.6 years (IQR, 3.9, 5.3), while the ABAS-II 6–21 was completed by 26 parents at a median age of 7.2 years (IQR, 6.7, 7.3) at follow-up. The ABAS-II 0–5 mean general adaptive composite score for the CDH cohort under 6 years of age was 98.7 ± 16.9 (IQR 92, 110), conceptual adaptive mean score was 100.0 ± 16.1 (IQR 94, 112), social adaptive mean score was 100.4 ± 16.5 (IQR 95, 112), and practical adaptive mean score was 98.2 ± 15.9 (IQR 92, 111). The ABAS-II 6–21 mean general adaptive composite score, conceptual adaptive score, social adaptive score, and practical adaptive score for those CDH children 6 years and older were 99.3 ± 17.2 (IQR 92, 109), 99.7 ± 16.8 (IQR 95, 112), 103.1 ± 17.3 (IQR 94, 115), and 96.9 ± 16.1 (IQR 88, 109), respectively. There was no difference in the mean main domains and

Table 2

CDH cohort main domain executive function, adaptive function, and behavioral outcomes compared with expected population normative data.

Instrument and domain	CDH-cohort	Expected normative sample mean ± SD	P-value	Proportion with abnormal scores
BRIEF-P (n = 38)				
Inhibitory Self Control Index	48.3 ± 13.1	50 ± 10	0.299	8/38 (21)
Emergent Metacognition Index	51.4 ± 15.7	50 ± 10	0.389	7/38 (18)
Global Executive Composite	49.8 ± 14.4	50 ± 10	0.909	8/38 (21)
BRIEF-2 (n = 25)				
Emotional Regulation Index	55.0 ± 12.0	50 ± 10	0.012	9/25 (36)
Cognitive Regulation Index	54.2 ± 11.7	50 ± 10	0.038	8/25 (32)
Global Executive Composite	55.1 ± 13.2	50 ± 10	0.010	10/25 (40)
ABAS-II 0–5 (n = 42)				
GAC	98.7 ± 16.9	100 ± 15	0.591	5/42 (13)
Conceptual	100.0 ± 16.1	100 ± 15	0.992	4/42 (10)
Social	100.0 ± 16.5	100 ± 15	0.859	3/42 (7)
Practical	98.2 ± 15.9	100 ± 15	0.454	7/42 (18)
ABAS-II 6–18 (n = 26)				
GAC	99.3 ± 17.2	100 ± 15	0.821	2/26 (8)
Conceptual	99.7 ± 16.8	100 ± 15	0.916	4/26 (15)
Social	103.0 ± 17.3	100 ± 15	0.304	3/26 (12)
Practical	98.6 ± 16.1	100 ± 15	0.283	3/26 (12)
CBCL 1.5–5 (n = 41)				
Attention problems	53.7 ± 7.0	50 ± 10	0.018	7/41 (17)
Attention deficits hyperactivity problems	52.5 ± 5.0	50 ± 10	0.104	4/41 (10)
Internalizing problems	46.0 ± 11.9	50 ± 10	0.009	8/41 (20)
Externalizing problems	44.8 ± 12.1	50 ± 10	0.001	5/41 (12)
Total problems	46.3 ± 12.5	50 ± 10	0.018	6/41 (15)
CBCL 6–18 (n = 22)				
Attention problems	56.5 ± 9.4	50 ± 10	0.001	6/22 (26)
Attention deficits hyperactivity problems	54.6 ± 6.8	50 ± 10	0.027	5/22 (22)
Internalizing problems	48.4 ± 11.4	50 ± 10	0.452	4/22 (17)
Externalizing problems	45.0 ± 10.6	50 ± 10	0.016	2/22 (9)
Total problems	47.2 ± 11.9	50 ± 10	0.175	3/22 (13)

Data presented as mean ± standard deviation or n (%) as appropriate. BRIEF-P, Behavior Rating Inventory of Executive Function Preschool; BRIEF-2, Behavior Rating Inventory of Executive Function 2nd Edition; ABAS-II 0–5, Adaptive Behavioral Assessment System 2nd Edition 0–5 years; ABAS-II 6–21, Adaptive Behavioral Assessment System 2nd Edition 6–21 years; CBCL 1.5–5, Child Behavior Checklist 1.5–5 years; CBCL 6–18, Child Behavior Checklist 6–18 years.

Boldface values highlight significant levels.

total scores of the CDH children compared to the expected normative mean scores (Table 1). Mean subscales scores and distributions were also similar to those of the normative population (Supplemental table 2).

Although mean values and distributions for the parent-reported ABAS scores were similar between our cohort and the normative data, the gestational age at delivery was more likely to be lower for CDH children with below average (worse, defined as $<1SD$) general adaptive composite (37 weeks [34, 39] vs. 39 weeks [39, 40], $P = 0.007$), social adaptive score (37 weeks [24, 38] vs. 39 weeks [34, 38], $P = 0.005$), and practical adaptive score (38 weeks [36, 39], vs. 39 weeks [39, 40], $P = 0.033$) scores than those with main domains scores within the average expected range. Lower social adaptive scores were also seen in CDH children who were more likely to have a public insurance (60 % vs. 12 %, $P = 0.026$). There was a trend towards lower social adaptive scores depending on race, with CDH children with lower social adaptive scores less likely to be white (50 % vs. 85 %, $P = 0.072$). Below average practical adaptive scores were also seen in CDH children who were more likely to require medical management of pulmonary hypertension (40 % vs. 9 %, $P = 0.024$). The multivariable logistic regression model for worse general adaptive composite score adjusted for Race/Ethnicity, maternal insurance, ECMO use, and age at initial enteral feeding indicated that higher gestational age at birth was associated with lower odds [$aOR = 0.53$ (0.28, 0.88)] of worse general adaptive composite.

3.5. CBCL ($n = 63$)

The median follow-up age for CDH children for which parents completed the CBCL 1.5–5 years ($n = 41$) and for which parents completed the CBCL 6–18 years ($n = 22$) was 4.53 years (IQR, 3.83, 5.34) and 7.27 (IQR, 6.79, 7.60), respectively. The mean internalizing score (46.0 ± 11.9 [IQR 37, 55]), externalizing score (44.8 ± 12.1 [IQR 35, 51]), and the total problem score (46.3 ± 12.5 [IQR 37, 53]) for the CBCL 1.5–5 CDH children were better than the expected means of the general population. The mean externalizing score (45.0 ± 10.6 [IQR 36, 51]) for CDH children for which parents completed the CBCL 6–18 was also better than expected ($P = 0.016$), the CBCL 6–18 internalizing (48.3 ± 11.4 [IQR 41, 57]) and total score (47.2 ± 11.9 [IQR 35, 56]) for our CDH cohort were similar when compared population norms. Independent of age at follow-up mean attention problem scores were significantly higher (worse) for the CDH children (CBCL 1.5–5, 53.7 ± 6.5 [IQR 50, 53], $P = 0.018$ and CBCL 6–18, 56.5 ± 9.4 [IQR 50, 59], $P = 0.001$). Attention deficits hyperactivity problem scores for school age CDH survivors were also significantly higher than the general population mean in this subdomain (54.6 ± 6.8 [IQR 50, 58], $P = 0.027$). Detailed subscales of the CBCL data are reported in Supplemental table 3.

Consistent with the BRIEF and ABAS results, race was also associated with abnormal ($\geq 1SD$) CBCL scores. CDH children with abnormal mean externalizing scores tended to be less likely to be white compared to those with average scores (57 % vs. 81 %, $P = 0.081$) while CDH survivors with total scores, and attention deficits hyperactivity problems with abnormal scores were less likely to be white (44 % vs. 84 %, $P = 0.030$ and 44 % vs. 84 %, $P = 0.009$). Additionally, abnormal total scores were found in CDH children who were more likely to have public insurance (44 % vs. 12 %, $P = 0.034$). Use of ECMO (44 % vs. 15 %, $P = 0.055$), later age initiation of enteral feeding (36 days [13, 40] vs. 16 days [9, 27], $P = 0.052$) and later age at reaching full enteral feeding (48 days [25, 58] vs. 29 days [16, 40], $P = 0.055$) were different between CDH survivors with abnormal total scores compared to those with average total scores. Analogously, abnormal internalizing scores were also more likely found in CDH children with delayed starting age of enteral feeding (26 days [15, 37] vs. 15 [9, 27], $P = 0.048$) and delayed age in reaching full enteral feeding (37 days [30, 58] vs. 28 days [16, 40], $P = 0.059$). Multivariable logistic regression model for abnormal total score adjusted for Race/Ethnicity, insurance, ECMO use, gestational age at delivery, and age at initial enteral feeding did not reveal

any statistically significant associations.

4. Discussion

In this study we characterize the executive, adaptive, and behavioral functioning of a large cohort of CDH survivors throughout preschool and early school age using standardized assessment. Our results demonstrate several important findings. First, although the majority of CDH survivors have age-appropriate executive function, CDH school age children have greater executive function difficulties in daily life than the normal population. We found significantly elevated mean scores and proportions with abnormal scores for the BRIEF-2 global executive composite and emotional regulation index. However, since grouping of higher-order cognitive and behavioral skills on the BRIEF is not perfect and subdomains are not independent from each other, we did not find a specific executive function impairment profile in this high-risk population. Normal executive functioning skills emerge during preschool years and develop with age, mirroring increased environmental and academic demands on goal-directed cognitive and behavioral skills. The younger age of children of the parent-reported BRIEF-P data might explain why we did not find differences in executive functioning in preschool CDH survivors. Second, as a group, CDH survivors have favorable adaptive functioning during preschool and early school age. CDH children performed well within the normal range, with the mean scores and distribution of scores of the entire cohort similar to the expected general population, suggesting that the acquisition of adaptive skills appears normal in most CDH survivors in the current cohort. Finally, children with CDH had better than expected or similar scores for main CBCL results (e.g., internalizing, externalizing and total problems) suggesting that most children with CDH appear to have no or only mild behavioral problems in school. We found, however, that preschool patients were more likely to have elevated scores for attention problems, and school age patients were more likely to have both attention and attention deficit hyperactivity problems. These findings perhaps indicate a progression of early difficulties with inattention to increased symptoms consistent with attention deficit/hyperactivity disorder by school age.

The current study adds to and expands the limited research on functional abilities in CDH survivors [23]. Our findings are consistent with recent data in adolescents with congenital malformations, including patients with CDH, reporting a distinctive delay in executive functioning with abnormal scores in main BRIEF domains including Behavior Regulation Index and Metacognition Index in survivors of critical neonatal illness [16]. However, the authors did not report outcomes separately for CDH survivors. Therefore, it is not possible to differentiate if the executive function deficits observed in their study were driven by the CDH patients or by survivors of the other included congenital anomalies. Nevertheless, the same group reported the outcomes of 40 school aged CDH survivors in a different study and showed that up to 40 % of these children were diagnosed with executive dysfunction confirming that this vulnerable population is at increased risk of adverse outcomes [24]. Peetsold et al. [25] previously studied neurobehavioral outcomes in 28 school age and adolescents CDH survivors and demonstrated that one in five of these patients had CBCL scores within the clinically significant range. Although comparing results from various studies is somewhat limited, as discrepant results may in part be explained by different methodology, age at assessment, selection criteria, and definitions of impairment, our study in combination with these previous investigations suggests that CDH survivors are at increased risks of impaired executive function and behavioral problems in certain domains. These developmental and neurobehavioral deficits may reduce functional independence, decrease social skills and peer to peer interaction, and increase behavioral dysfunction. In turn, these deficits, may be associated with poor school performance and less successful social and peer interaction as children mature. As higher neurocognitive and behavioral functioning gradually develop with age, it

might not be surprising that we did find noticeable abnormalities in preschool CDH survivors.

The pathophysiological mechanisms underlying neurologic injury in CDH are diverse, likely multifactorial, and not yet fully understood [1,2]. We identified several patient-specific and management related risk factors that in some, but not all, neurobehavioral domains negatively influence outcome, namely prematurity, surrogate markers of disease severity, ethnicity (non-white race), and public insurance status. The risks of prematurity and adverse neurological outcomes in children with and without CDH are well documented [5,14]. It is therefore not surprising that children with CDH who were born earlier were more likely to have abnormal scores among several ABAS domains than those born closer to term. While prematurity is also associated with executive dysfunction and neurobehavioral problems in the general population [26,27], the nonsignificant differences in BRIEF and CBCL scores may be related to insufficient power to detect small effect sizes between term and premature children with CDH in our cohort. CDH survivors who required medication for the management of pulmonary hypotension and/or ECMO were more likely to have lower scores across several BRIEF, ABAS, and CBCL domains. Delays in the ability to start enteral feedings were associated with lower CBCL scores. Whether this management-related variable itself puts children with CDH at risk for functional and behavioral problems or reflects a more severe primary disease influencing neurobehavioral outcome remains a topic of debate. The findings that non-white race and the presence of public insurance are associated with higher risks of abnormal scores in various domains of executive, adaptive, and behavioral functioning is consistent with growing evidence that innate patient and environmental factors are equally as important to consider in neurobehavioral outcomes [2]. Of note, the impact of ethnicity, race, and need for public insurance, on neurophysiological impairment is not necessarily specific to children with CDH [2,28–30]. However, early identification of these environmental risk factors may allow for improved resource allocation and support for these families and in turn improve neurobehavioral functioning, [31].

We were unable to identify statistically significant associations between any other clinical variables listed in Table 1 and neurobehavioral outcomes. It is possible, however, that additional risks factors could be identified in a larger sample. As enrolling for this study occurred during the Covid pandemic, concurrent onsite evaluation of school-age neurocognitive outcomes using the Wechsler Preschool and Primary Scale of Intelligence, the Developmental Test of Visual Motor Integration, or the Woodcock-Johnson Psychoeducational Battery-Revised was not feasible. Future studies should investigate whether CDH survivors with impairment in higher-level cognitive functioning and academic achievements are at increased risk of adverse neurobehavioral outcomes – an association frequently seen in long-term survivors of complex congenital heart disease [32–35].

Although our data provide additional information on executive function, adaptive functioning, and behavioral outcomes in preschool and school age children with CDH, our study has certain limitations and caveats that need to be considered. First, our results need to be interpreted in the setting of a retrospective study design, including inherent information and selection bias. Second, even though our study is one of largest studies on executive function, adaptive function and behavioral outcomes in CDH to date, risk factor analysis remains limited by statistical power. Third, detailed evaluation of socioeconomic and parental educational status was not readily available across the sample. Given the association of insurance status (which can be viewed as surrogate for socioeconomic and parental educational status) and the increased risk of adverse outcome, this information will have to be included in future data collection and analysis. It is also important to note that patients with public insurance were less likely to participate. Given that, our results show an association between public insurance and increased risks of adverse outcomes, our study population may not adequately represent the CDH population seen at our institution. Finally, outcome

measures were based on parent-reports, which can be biased and may over or underestimate a child's capacity of independence and self-regulation. In addition, because not all parents completed all the questionnaires provided, the possibility of selection and response bias remains. We therefore recommend that future work also collect teacher-report data to complement the caregiver assessment.

Despite these limitations, our study has important implications for clinical practice. Neurodevelopmental follow up in CDH survivors has been suggested as a practice guideline by the American Academy of Pediatrics [36], but there are no guidelines to assess specific areas of functional and neurobehavioral vulnerabilities. Our work suggests that follow-up should include detailed assessment of executive function and behavioral function to identify patients with subtle difficulties achieving age-appropriate self-care skills and independence. Without focused evaluations, particular longer-term follow-up these impairments may go undiagnosed and delay the timely implementation of appropriate therapies to optimize psychosocial adjustments and behavioral functioning in these children.

In summary, we found that the majority of preschool and school age children with CDH have age-appropriate executive and behavioral functioning, but lower executive function and attention scores are more common in this vulnerable population than the general population. Although these results may provide some important reassurance for parents, targeted screening and improved understanding of the specific patterns of cognitive, functional, and behavioral problems in these patients are needed to provide individualized support to promote optimal developmental outcomes. Additional research is warranted to further examine the relationship between parental reporting of executive and behavioral functioning, and the child's actual academic achievement and neurophysiological outcome. To address this need longer-term assessment (8–13 years) of our CDH population is currently underway including in-person standardized evaluations of the child's overall functional status by our neurophysiologists.

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CRediT authorship contribution statement

Enrico Danzer: Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Investigation. **Jane E. Schreiber:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Casey Hoffman:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Lenny Mathew:** Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – review & editing. **Sabrina J. Flohr:** Data curation, Investigation, Methodology, Validation, Writing – review & editing. **Elizabeth Eppley:** Data curation, Investigation, Methodology, Writing – review & editing, Validation. **Sierra D. Land:** Data curation, Methodology, Validation, Writing – review & editing. **Lisa Herkert:** Investigation, Writing – review & editing. **Natalie E. Rintoul:** Investigation, Writing – review & editing. **N. Scott Adzick:** Investigation, Writing – review & editing. **Holly L. Hedrick:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

None of the authors has any conflict of interest to disclose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.earlhumdev.2023.105914>.

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